

PROJECT PROPOSAL

Film Festival Scheduling + Venue Management

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Objective

This project proposes a model-driven engineering (MDE) solution to plan and operate a film festival by jointly addressing screening schedules and venue management. The goal is to assign films with durations, categories, audience tags, directors, and optional Q&A sessions; to venue screens with specific seat capacities, equipment, and availability windows; producing a feasible, optimized schedule. The system will integrate a constraint solver to compute assignments, and will generate two principal textual artifacts: a public-facing daily program and CSV exports for a ticketing system. The approach is explicitly metamodel-driven and uses transformations: model-to-model (M2M) for normalization and solver model creation, text-to-model (T2M) to ingest solver results, and model-to-text (M2T) to publish programs and exports.

The solution contains three core domains and one integration domain. In the Festival domain, curated films must be shown at least once (often multiple times), each with a fixed duration and possibly an attached Q&A session that must be scheduled right before or after a screening, within festival days and time windows. In the Venues domain, theaters comprise of one or more screens, each with seat capacity, seat maps, equipment type, and availability windows. The Overlap domain captures the assignment problem: each screening must be assigned a start time and a screen without overlaps that violate screen availability, turnover buffers, or equipment requirements, while considering soft preferences such as grouping by audience tags, balancing categories across slots, and avoiding personnel conflicts.

The practical problem is to compute a feasible schedule that respects hard constraints (e.g., non-overlap, equipment, capacity, availability, Q&A adjacency) and optimizes soft goals (e.g., utilization, audience-fit, category spread, minimizing high-demand conflicts). Once computed, the schedule must be presented as a daily program and exported through ticketing CSVs.

Models

- Models and metamodels

The system will use four metamodels, all defined in EMF/Ecore with constraints where appropriate.

- Metamodels (MMs):
 1. FestivalMM (Festival Domain)

- Film with title, duration, category, audienceTags, premiereFlag, expectedDemand (Low/Med/High), director(s).
 - Person with name and role (Director, Guest).
 - Category and AudienceTag as controlled vocabularies.
 - QASession with duration, participants (subset of Person), and adjacencyPolicy (e.g., immediately after screening, within N minutes, same screen or near-venue).
 - Day and TimeWindow definitions for curatorial blocks and constraints.
2. VenueMM (Venues Domain)
 - Venue with name, address, and a set of Screens.
 - Screen with id, seatCapacity, seatMap (optional fine-grained layout), equipment (ProjectorType, AudioConfig), and turnoverBuffer.
 - Availability with open/close times and non-operation windows (maintenance, special preparations).
 - Equipment and AccessibilityFeature enumerations.
 3. ScheduleMM (Overlap Assignment Domain)
 - Screening as an assignment of a Film to a Screen with startTime, endTime (derived), and status (Planned/Scheduled/Locked).
 - Turnover and Cleanup as derived attributes; hardBoundaries for curatorial mandates.
 - QAAssignment tied to a Screening, with startTime, location (screen or adjacent hall), and participants.
 - ObjectiveWeights containing preference weights (e.g., seat utilization, category balance, tag clustering, conflict avoidance).
 - Validation constraints such as non-overlapping screenings on the same screen, capacity $\geq$ expected demand, equipmentMatch(Film, Screen), Q&A adjacency, and schedule respected.
 4. ConstraintMM (Solver Integration Domain)
 - Variable, Domain, Constraint (Hard/Soft with weight), and Objective (maximize weighted sum).
 - TimeDomain splits time into discrete slots, and each Screen is a Resource with capacity=1 and sequencing to prevent overlapping screenings.
 - Mappings to identify each Film candidate Screen/Timeslots, required equipment predicates, buffer constraints, and Q&A adjacency as precedence/spacing constraints.
 - ConstraintMM serves as a intermediate model for solvers. It enables usage of different solvers without modifying the domain models.

Domains, MMs, and their relations.

- There are four domains:
 1. Festival (FestivalMM)
 2. Venues (VenueMM)
 3. Solver integration (ConstraintMM)
 4. Overlap/Scheduling (ScheduleMM)

- Relations:
 - FestivalMM and VenueMM are independent foundational models.
 - ScheduleMM references instances from both FestivalMM and VenueMM to express assignments; it does not duplicate film or screen data.
 - ConstraintMM is created by an M2M that consumes FestivalMM and VenueMM to be used by the solver.
 - The solver produces text results that will be converted via T2M into ScheduleMM.
 - Publication and ticketing generators consume FestivalMM, VenueMM, and ScheduleMM.
 - One Festival model per event; one Venue model per festival.
 - At least one Schedule models (maybe different scenarios) can be used for what-if planning.
 - One derived Constraint model per solver run.

Tool integration

The system integrates:

- A constraint solver, preferably something in Python, will be integrated into the system, with relevant packages and libraries.
- Ticketing system: CSV-based export with specified columns (screening id, film title, date, start, end, venue, screen, capacity, language/subtitle, price tier).

Model compliance strategy:

- M2T will generate a text based file to be used by the solver, referencing ConstraintMM to ensuring correctness.
- Solver output will be in a similar text-based format. A T2M parser will create ScheduleMM.
- Ticketing CSV structure is documented; an M2T ensures required columns, escaping, and data types are correct and the CSV is exported.

Model creation and update

Initial creation:

- FestivalMM is manually created. Information related to films, durations, categories, tags, expected demand, and Q&A details are entered or imported from curated spreadsheets.
- VenueMM is also manually created: screens, capacities, equipment, and availability/blackouts are loaded from a venue spreadsheet or manually entered.

Updates:

- ConstraintMM is M2M derived from FestivalMM and VenueMM.
- ScheduleMM is T2M derived from the solver output.

- Program changes (film added, duration changed, Q&A participants updated), venue operations updates (screen maintenance, equipment change), or policy changes (new buffers or preference weights) mark affected elements as dirty.
- A model listener layer tracks changes and triggers the solve pipeline: listener trigger → M2M to ConstraintMM → M2T to solver input → solver run → T2M back to ScheduleMM.

Transformations

The project will deliver the following core transformations, illustrated in Figure 1:

- M2M: Festival+Venue → ConstraintMM
 - Collects feasible Screen/Timeslot candidates per Film given availability windows, equipment predicates, and buffers.
 - Encodes non-overlap per Screen, Q&A adjacency/precedence, category quotas or spread constraints, and soft preferences (e.g., seat utilization, layout preference, conflict avoidance).
 - Produces an objective aggregation (weighted sum) parameterized by ScheduleMM.ObjectiveWeights.
- M2T: ConstraintMM → Solver Input
 - Depending on the tools used, appropriate text-based results will be provided to the solver.
- T2M: Solver Output → ScheduleMM
 - Parses variable assignments (screen, start time slot) into Screening instances and computes end times from Film.duration plus buffers.
- M2T: Ticketing CSV Export
 - Generates CSV files with the exact schema required by the ticketing system.

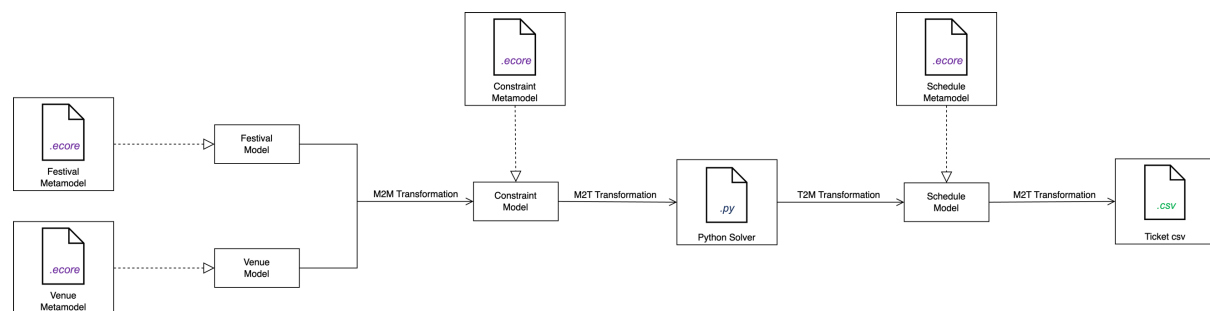


Figure 1: Overview of the transformations to be implemented

Combining transformations

Pipeline to solve scheduling:

1. Validate FestivalMM and VenueMM to ensure all required attributes are present.
2. M2M: Build ConstraintMM from FestivalMM and VenueMM.
3. M2T: Export model to solver input.
4. Execute solver with measures to ensure reproducibility.
5. T2M: Parse solver output into ScheduleMM.
6. M2T: Produce ticketing CSV files for the corresponding timeframe